

Report - Version 1: Simple Single Neuron

Dataset	Task	Train acc.	Test acc.
Fashion-MNIST	0 vs 1	98.97%	99.15%

Introduction

Version 1 is the first working version of the Building Neural Networks project. The goal is to build the smallest possible neural model from scratch, without using a machine-learning library for the model itself.

The architecture is intentionally minimal: 784 normalized pixel values are combined by a single neuron, a sigmoid turns the raw score into a probability, and gradient descent learns the weights and bias.

Goal of Version 1

The main goal is to understand the learning loop of a neural network before introducing multiple classes, hidden layers, mini-batches, or deeper architectures.

Data and Preparation

The project uses the Fashion-MNIST CSV dataset. Each 28×28 grayscale image is represented by 784 pixel values and one label.

Version 1 keeps only two classes: 0 = T-shirt/top and 1 = Trouser. This creates a simple binary classification problem. Pixel values are normalized from 0-255 to 0-1 by dividing by 255.

Architecture

The flow of Version 1 is:

784 pixels -> weighted sum -> sigmoid -> probability -> binary prediction

Single Neuron

The neuron computes a weighted sum of the inputs plus one bias. With 784 pixels, the model therefore learns 784 weights and one bias.

$$z = X @ w + b$$

Sigmoid Activation

The sigmoid converts the raw neuron score into a value between 0 and 1. This value is interpreted as the probability of class 1 (Trouser). A score near 0 means the model favors T-shirt/top; a score near 1 means it favors Trouser.

Loss and Gradient Descent

Binary cross-entropy measures how well the predicted probabilities match the true binary labels. Gradient descent then adjusts the weights and bias in the direction that reduces this loss.

The implementation uses a learning rate of 0.1. Training repeats until the improvement in loss becomes smaller than $1e-6$.

Training Result

Weights start at zero and the bias starts at 0.0. The final training loss is approximately 0.0305. On the binary training subset, the model reaches 98.97% accuracy.

Evaluation

The separate Fashion-MNIST test CSV also contains labels. Evaluation is therefore performed directly on the test subset containing only classes 0 and 1.

A probability threshold of 0.5 converts the sigmoid output into a binary class. The resulting test accuracy is 99.15%. This is an evaluation result, not a Kaggle submission score.

Metric	Value
Training loss	0.0305
Training accuracy	98.97%
Test accuracy	99.15%

What Version 1 Teaches

Version 1 exposes the complete learning loop of a single neuron: initialize parameters, compute a forward pass, measure loss, compute gradients, update parameters, repeat, and finally convert probabilities into classes.

It also shows why a very simple linear model can perform extremely well when the chosen two-class problem is easy to separate.

Limitations

Version 1 deliberately supports only two Fashion-MNIST classes, has no hidden layer, uses full-batch gradient descent, and does not yet create a full 10-class classifier. The training loop also has no explicit maximum-epoch safety limit.

These are intentional constraints that keep the first version small and define the next development step.

Conclusion

Version 1 successfully builds a working image classifier from scratch with only NumPy for the learning logic. A single neuron reaches 98.97% training accuracy and 99.15% test accuracy on T-shirt/top versus Trouser.

The next natural step is Version 2: replace the single sigmoid output with 10 outputs and softmax so the model can classify all Fashion-MNIST classes.